

ICE SQLALCHEMY VALIDATION

Migration Validation

Cost-Benefit Analysis

June 2026

Executive summary

Less manual checking, lower migration risk, faster sign-off.

Time per check

≈ 0.03 s

Automated, vs 80.5 min done by hand

Saved per year

21 h

About £1.6k of engineer time

Setup cost

£600

One-time, 8 h to onboard

Why it matters

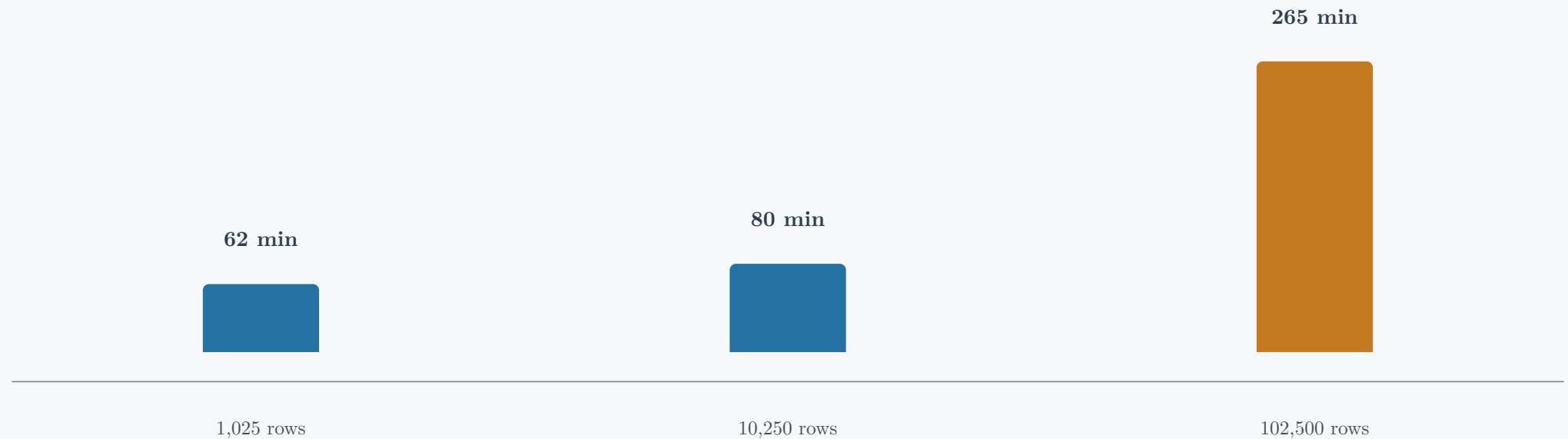
The real value is avoiding expensive late surprises, not only saving time.

The risk today

Real incidents were found by customers after go-live, not by internal checks.

Manual effort grows with data size

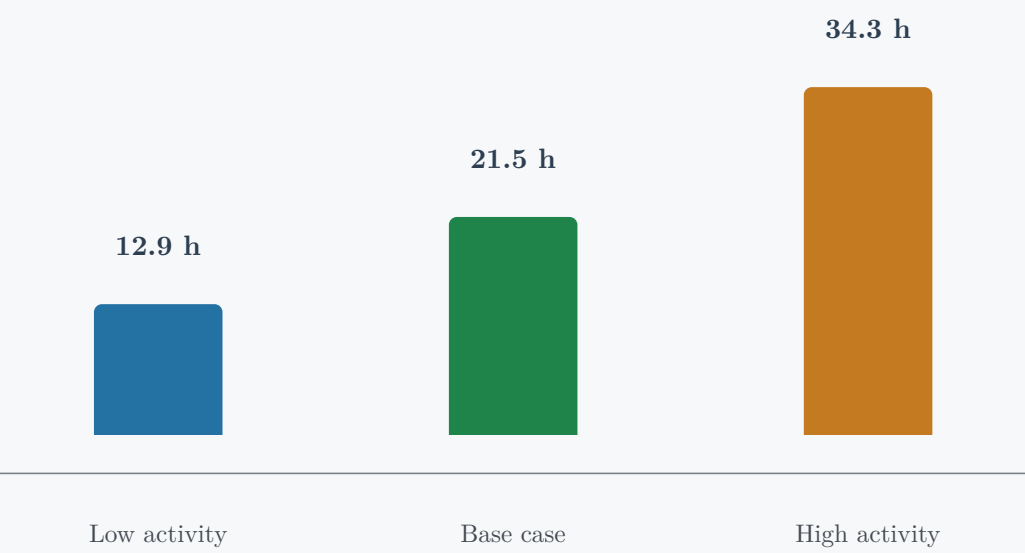
Automated checks stay under one second at every size tested.



Automated check: **0.03 – 0.56 seconds** across all three sizes — effectively a flat line near zero.

Annual time saved

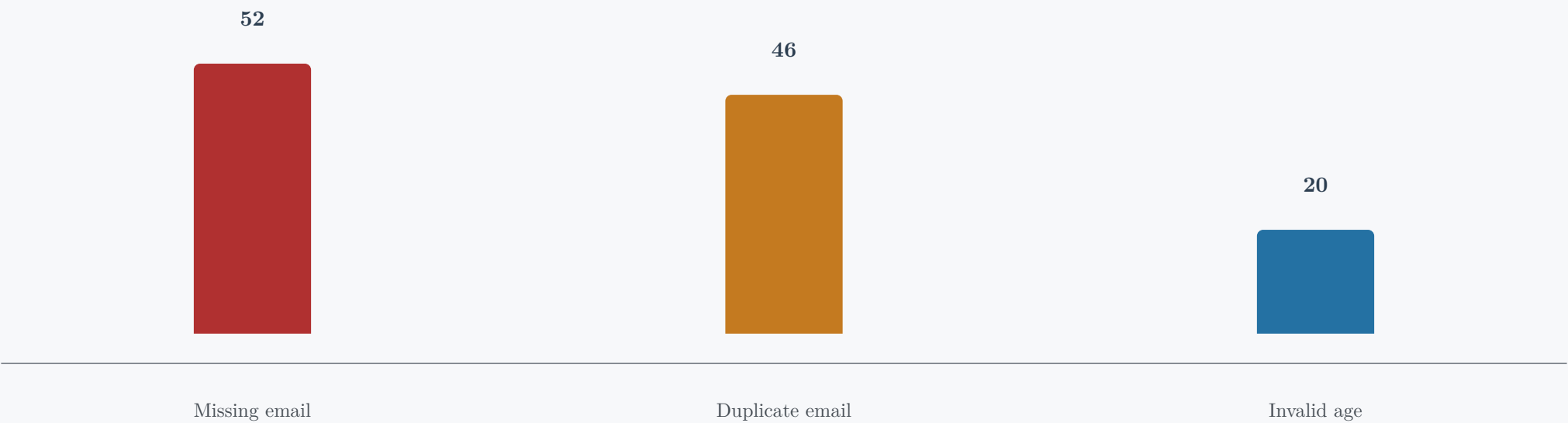
Low, base, and high activity levels for migration checks.



Scenario	Hours / yr	Value / yr
Low	12.9	£968
Base	21.5	£1.6k
High	34.3	£2.6k

Defects found in validation

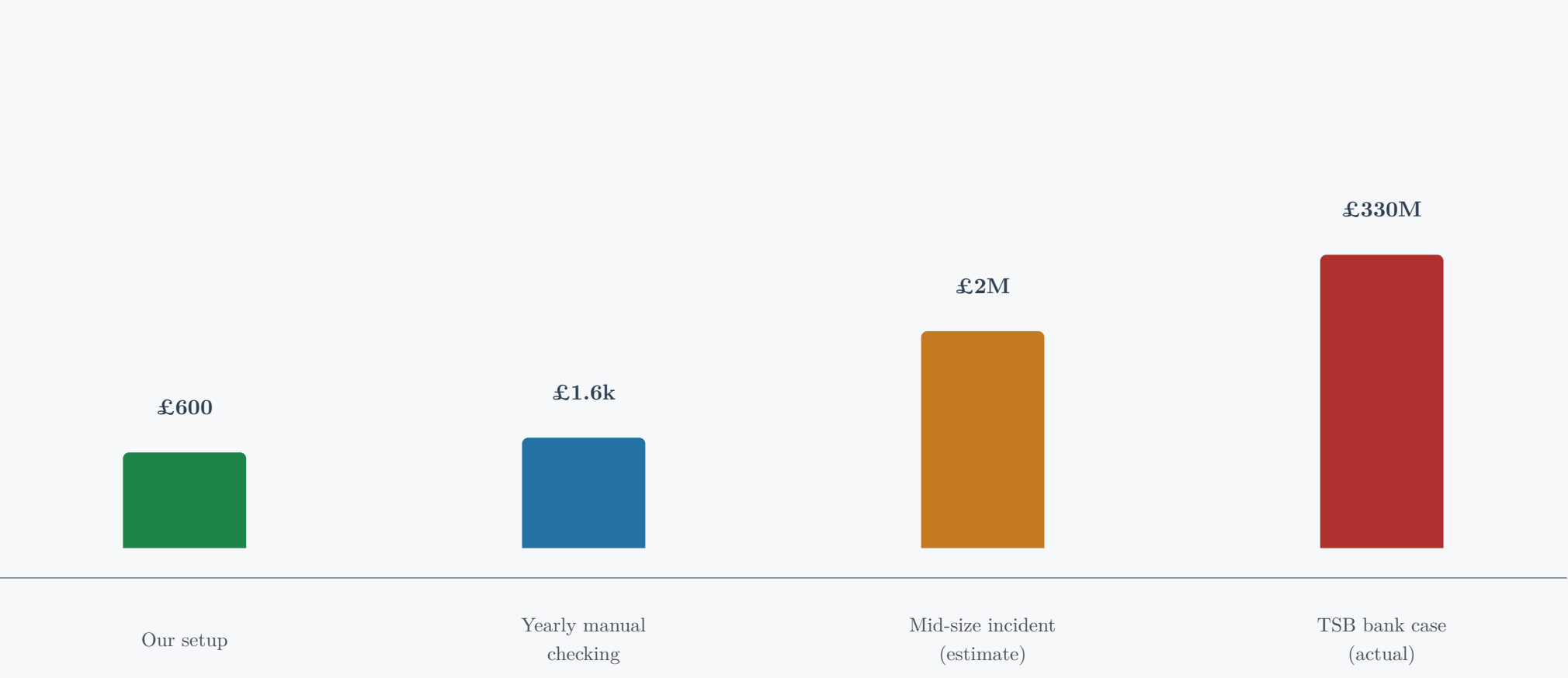
Sample run on 1,025 rows with deliberately dirty data.



118 bad records would have passed a row-count check unnoticed. These are exactly the silent errors behind documented real incidents.

Cost of a missed defect

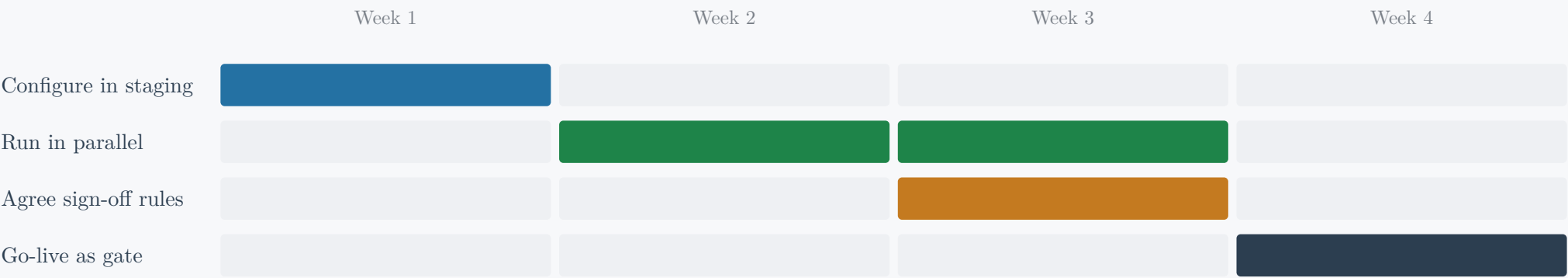
Setup cost compared with manual checking and documented incidents (log scale).



Setup is a rounding error next to a single real incident. The TSB migration cost £330M; our control costs £600 once.

Rollout plan

Four weeks to adopt alongside the current process.



Phase	Activity	Owner
Weeks 1	Configure checks and capture a baseline	Data engineer
Weeks 2–3	Run beside the manual process, compare results	QA + PM
Week 4	Make it a required pre-release gate	Release manager

Recommendation

Approve now

- 8 h budget for implementation
- Pilot on the next migration cycle
- Add a pass / fail gate before release

Expected outcome

- Lower chance of post-go-live data incidents
- Faster, evidence-based release confidence
- Clear go / no-go signal for stakeholders

Bottom line: a £600 one-time control that saves £1.6k a year and guards against incidents measured in millions. It improves predictability with almost no added process overhead.

References

- `examples/_bench_timing.py` — measured manual vs automated timings
- `examples/validate_dirty_database.py` — real defect categories and counts
- `src/` — the validation engine used in the benchmark
- Public migration incident reports — TSB, insurance, MySQL → PostgreSQL, OpenAI, TUI